

Introductions.

JS/Linux

Console = Terminal.

whoami = Computer name.

```
Welcome to JS/Linux (x86_64)

Use 'vlogin username' to connect to your account.
You can create a new account at https://vfsync.org/signup .
Use 'export file filename' to export a file to your computer.
Imported files are written to the home directory.

localhost:~# whoami
root
localhost:~# date
Mon Aug 24 10:51:56 UTC 2026
localhost:~#
```

Linux Package Management.

- Debian/Ubuntu

.deb = Package for debian/ubuntu. Installed, updated and removed

- Management tools.

- APT = high-level, works with only repos (apt-get, apt-cache)
- dpkg = low-level, works with standalone .deb file (CD/DVD, disk)
- Aptitude = layer over apt + dpkg → automates tasks

Canonical Repos

- main = free, Ubuntu-supported
- restricted = not 100% free, still supported
- Universe = free software, Unsupported
- Multiverse = non-free, Unsupport

System-changing commands → require sudo.

APT

Command	Function
apt-get update	Refreshes package list
apt-get install <pkg>	Downloads + installs
apt-get remove --purge <pkg>	Removes package + config
apt-get upgrade	Updates all installed packages
apt-get dist-upgrade	Full system update (may remove pkgs)
apt-get -f install	Repairs broken dependencies
apt-get clean	Clears download cache

Sudo apt-get update;
sudo apt-get upgrade

Always update before upgrade.



DPKG

Command	Function
<code>dpkg -i file.deb</code>	Installs a .deb file
<code>dpkg -l <pkg></code>	Checks installed version
<code>dpkg -L <pkg></code>	Lists files installed by package
<code>dpkg -s <pkg></code>	Status / details
<code>dpkg -r <pkg></code>	Removes (keeps config)

low-level tool behind apt
works with standalone .deb files.

`Apt -P <pkg> → dpkg -i file.deb.`

Aptitude

Command	Function
<code>aptitude search <word></code>	Searches by keyword
<code>aptitude update</code>	Refreshes indexes
<code>aptitude dist-upgrade</code>	Updates all, adds/removes as needed
<code>aptitude install <pkg></code>	Installs
<code>aptitude remove <pkg></code>	Removes

Curses UI (no options)
CTRL + T menu · Q → Quit
auto-updates before install upgrade.



`-y` auto - confirms → aptitude has no undo

```
user@ubuntu:~$ aptitude search nginx
p  nginx                - small, powerful, scalable web/proxy server
p  nginx-common          - small, powerful, scalable web/proxy server
p  nginx-doc             - nginx documentation

user@ubuntu:~$ sudo aptitude install nginx
The following NEW packages will be installed:
  nginx nginx-common
0 packages upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 600 kB of archives. After unpacking 2,000 kB will be used.
Do you want to continue? [Y/n/?] Y
Writing extended state information... Done
Get: 1 http://archive.ubuntu.com/ubuntu main/nginx-common amd64 1.18.0 [35.5 kB]
Get: 2 http://archive.ubuntu.com/ubuntu main/nginx amd64 1.18.0 [564.5 kB]
Fetched 600 kB in 1s (600 kB/s)
[... installation output continues ...]
```

PART II - RED HAT

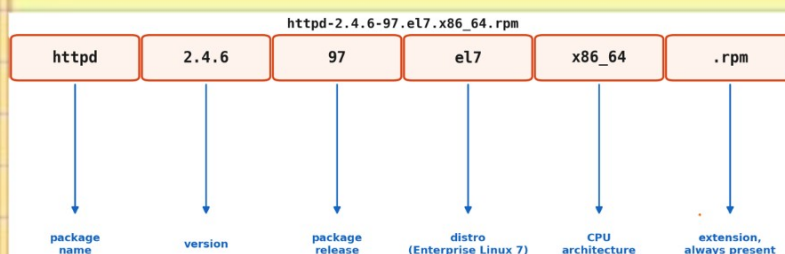
RPM vs YUM

RHEL = set of .rpm packages + linux kernel

RPM	YUM
Installs / queries .rpm files	Manages packages + repos
Does NOT resolve dependencies	DOES resolve dependencies

Package = files + metadata (scripts, configs, libs, docs)

Packages Name Anatomy



- Dependency = another package required to install/run this one

- Package DB: `var/lib/rpm` → tracks versions, file, checksums, deps.

RPM - Core Commands

Command	Function
<code>rpm -qa</code>	Lists all installed packages
<code>rpm -q <pkg></code>	Checks if installed
<code>rpm -qi <pkg></code>	Package info
<code>rpm -ql <pkg></code>	Lists files in package
<code>rpm -qf <file></code>	Finds owning package of a file
<code>rpm -ivh <pkg>.rpm</code>	Installs (-v verbose, -h progress)
<code>rpm -Uvh <pkg>.rpm</code>	Updates, or installs if not present
<code>rpm -ev <pkg></code>	Removes (fails if something depends on it)
<code>rpm -K <pkg>.rpm</code>	Verifies GPG signature + integrity



--nodeps ignores missing deps → not recommended

`rpm -V` checks installed vs DB - flags size/perm/checksum/owner changes.

YUM - core commands

YUM = front-end over rpm, resolves dependencies automatically

config: `/etc/yum.conf` • repos in `/etc/yum.repos.d/* .repo`

Command	Function
<code>yum install <pkg></code>	Installs (asks confirmation)
<code>yum -y install <pkg></code>	Installs without asking
<code>yum update</code>	Updates all installed packages
<code>yum remove <pkg></code>	Removes package + dependents
<code>yum search <word></code>	Searches name/description
<code>yum provides <file></code>	Finds package that owns a file
<code>yum info <pkg></code>	Shows package details
<code>yum history list</code>	Shows recent transactions

```
[user@rhel ~]$ yum history list
Loaded plugins: fastestmirror
ID      | Login user          | Date and time      | Action(s)      | Altered
-----|-----|-----|-----|-----
4       | admin <admin>       | 2026-08-25 09:00   | Install        | 1
3       | admin <admin>       | 2026-08-24 14:30   | Erase          | 2
2       | admin <admin>       | 2026-08-23 11:15   | Update         | 15
history list
```

```
[user@rhel ~]$ yum provides /etc/httpd/conf/httpd.conf
Loaded plugins: fastestmirror
httpd-2.4.6-97.el7.x86_64 : Apache HTTP Server
Repo                        : base
Matched from:
Filename                   : /etc/httpd/conf/httpd.conf
```

```
[user@rhel ~]$ rpm -qi httpd
Name       : httpd
Version    : 2.4.6
Release    : 97.el7
Architecture: x86_64
Install Date: Tue 25 Aug 2026 09:00:00 AM CST
Group      : System Environment/Daemons
Size       : 9400000
License    : ASL 2.0
Signature  : RSA/SHA256, Wed 15 Jul 2026, Key ID 24c6a8a7f4a80eb5
Source RPM : httpd-2.4.6-97.el7.src.rpm
Build Host : x86-01.bsys.centos.org
Relocations: (not relocatable)
Summary    : Apache HTTP Server
Description:
The Apache HTTP Server is a powerful, efficient, and extensible
web server.
```